

JPA vs. Hibernate vs. Spring Data JPA

How the specification, the engine, and the convenience layer differ

Aspect	JPA	Hibernate	Spring Data JPA
Nature	A specification — a contract of rules, no runnable code	A framework — the actual engine that fulfils the JPA contract	A convenience layer built above a JPA provider
Core role	Standardises how Java objects should map to relational data	Executes that mapping, generates SQL, manages entity state	Removes hand-written repository code via auto-generated interfaces
Governing body	Java Community Process (JSR 338)	Hibernate ORM team (Red Hat)	Spring Framework team (VMware/Broadcom)
Runs on its own?	No — it is only interfaces and annotations	Yes — can run independent of the JPA spec if needed	No — always sits on top of a JPA implementation
Developer effort	High — manual EntityManager handling	Moderate — lighter than raw JDBC, still explicit	Low — interface declarations replace most logic
Built-in CRUD repositories	Absent	Absent	Present — JpaRepository, CrudRepository, etc.
Query mechanisms	JPQL	JPQL, HQL, and native SQL	Method-name derived queries, JPQL, native SQL
Dependency chain	None — it is the baseline standard	Depends on the JPA API contract	Depends on both JPA and a provider such as Hibernate